

Tran Hoang Kien

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ABOUT ME

Ho Chi Minh University of Technology - VNU HCM

Bachelor of Engineering in Electronics and Telecommunications (Fourth-year student)

2022 – Present

- A reliable and patient individual with strong problem-solving and planning skills.
- Passionate about electronics and embedded systems.
- Seeking opportunities to develop expertise in embedded programming.

SKILL

Technical Skills: C, MCU drivers (UART, SPI, I2C, CAN), Real-Time OS (FreeRTOS), Hardware design, USB stack, TCP/IP networking, CubeSat Space Protocol, Debugging with Logic Analyzer & Oscilloscope.

Software Tools: Altium Design, STM32cubeIDE, VS Code, GitHub, ESP32-IDF, MPLAB X IDE.

Embedded Platforms: AVR, STM32 (ARM Cortex M3), ESP32, Raspberry Pi, SAMV71.

Soft skills: Time Management, Teamwork, Problem solving, Critical Thinking, Research Skills.

EXPERIENCE

Embedded Systems Intern

Student Research Lab – Ho Chi Minh University of Technology | 2025

Firmware Update over UART

(March 2026 – May 2026)

Description: Developed a production-style firmware update solution for Microchip **SAMV71**, enabling reliable in-field application upgrades over UART. Implemented a custom packet-based protocol with CRC protection and a Python host tool for firmware flashing, verification, and application control.

Responsibilities & Achievements:

- Designed the firmware update architecture supporting flash erase/write, memory readback, and jumping to the application.
- Implemented a robust UART communication protocol with packet framing, CRC32 validation and ACK/NACK flow control.

CubeSat Space Protocol (CSP) over CAN & KISS UART

(September 2025 – November 2025)

Description: Integrated the open-source CubeSat Space Protocol (CSP), an embedded communication protocol stack supporting packet routing, into Linux and STM32 platforms over CAN and KISS UART, enabling reliable command and file transfer across intermediate nodes.

Responsibilities & Achievements:

- Developed Linux-side C applications to transmit commands and files over CSP via KISS UART, integrating **Python** scripts for file handling and data streaming.
- Ported and integrated the open-source CSP stack into **STM32** firmware running FreeRTOS with CAN and UART interfaces.
- Configured CSP routing tables to enable multi-hop packet routing.
- Debugged and resolved multi-hop communication issues, ensuring reliable data transfer across Linux ↔ STM32 (intermediate node) ↔ STM32 under real-time constraints.

High-Speed USB File Transfer over CDC

(August 2025 – September 2025)

Description: Bidirectional **USB CDC** file transfer system with custom chunk-based frames; data buffered in STM32 SDRAM, **Python** on laptop handles chunking and file reassembly.

Responsibilities & Achievements:

- Developed **STM32** firmware in **FreeRTOS** and **Python scripts** on laptop for chunk-based file transfer (SOF, SEQ, LENGTH, DATA, EOF).
- Buffered chunks in STM32 **SDRAM via FMC** to reconstruct files; supported sending files back to laptop for reassembly.
- Achieved stable high-throughput bidirectional transfer in lab tests.

Embedded TCP/IP over USB ECM

(June 2025 – August 2025)

Description: Implemented STM32 as a virtual Ethernet adapter on host PC, integrating FreeRTOS+TCP stack based on TCP/IP for lightweight embedded networking with a minimal HTTP server.

Responsibilities & Achievements:

- Enabled **STM32** to be recognized as a **virtual Ethernet adapter** on Linux host PC.
- Integrated **TCP/IP-based FreeRTOS+TCP stack** with lightweight **networking** tasks, including a minimal **HTTP server**.
- Validated end-to-end stack functionality in Windows simulator (ping, TCP echo client, HTTP requests).

USB Composite (HID and CDC)

(April 2025 – June 2025)

Description: Developed a custom USB composite device combining HID and CDC classes on STM32 without using CubeMX/IOC, enabling simultaneous HID and CDC operation.

Responsibilities & Achievements:

- **Manually merged HID and CDC classes** (descriptors, configuration, driver code) on STM32.
- Configured endpoints for concurrent HID + CDC functionality.
- Verified reliable string transmission via HID and interactive console over CDC.

ACADEMIC PROJECTS

Handheld Calculator

- Version 1 (**Raspberry Pi**): Built a calculator receiving keypad input and displaying results on a laptop; implemented arithmetic and equation-solving algorithms in C with **multithreading**.
- Version 2 (**STM32**): Ported the system to STM32 using **FreeRTOS tasks** for parallel execution of **Halley and Secant root-finding methods**, achieving faster computation.

ESP32 BLE Sensor

- Developed BLE application on ESP32 using **ESP-IDF** to read DHT11 temperature and humidity sensor and send data to a mobile app.
- Implemented BLE advertising and **notification payloads**, **GPIO wake-up interrupts**, and **power state management** for low-power operation.

Alarm Clock with Temperature & Humidity (STM32)

- Developed an alarm clock reading DHT11 & DS3231, displaying information on TFT screen; users **set time/alarm** via rotary encoder.
- Programmed STM32 peripherals (GPIO, I2C), implemented buzzer alarm logic, and **stored configuration in Flash**.
- **Designed custom PCB** for compact, stable system operation.

ACTIVITIES

Ho Chi Minh University of Technology Student Research Lab – PAY IT FORWARD

Active member

- Familiar with ARM Microcontroller with peripherals (UART, I2C,...).
- Trained basic C programming for second-year students mentored career paths in Engineering for 30 students and prepared assessments for new members.

EDUCATION

GPA: 7.66.

TOEIC: 710.

Relevant Coursework: Embedded Systems Design, Embedded Programming (Implemented I2C slave using STM32 LL libraries), Microprocessors (Implemented UART and SPI in Assembly), Applied Electronics.